Jiayi Chen

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Analytics Engineer | Supply Chain, Operations & AI Automation

SUMMARY

Analytics engineer with a finance foundation who turns messy operational data into decisions, recovered dollars, and AI-automated workflows. Builds and owns production ETL pipelines, trusted data models, and self-service analytics across supply chain, manufacturing, and healthcare. Core tools: SQL, Python, dbt, Databricks, Tableau, and Power BI.

TECHNICAL SKILLS

Languages & Query: SQL, Python (Pandas, NumPy, PySpark), R, Stata, MATLAB, JavaScript

Data Engineering & Modeling: Databricks, dbt, Apache Airflow, ETL/ELT pipelines, data modeling, workflow orchestration, data quality monitoring, SPC, root cause analysis, SAP ERP, MongoDB, NoSQL, EDI

BI & Visualization: Tableau, Power BI, JMP Live, dashboard automation, self-service analytics

Domain & Methods: Supply-chain and Logistics analytics, Git, GitHub, Jira, statistics, econometrics, Lean Six Sigma

EXPERIENCE

TCI Biotech USA LLC *American Fork, UT*

Data & AI Analytics Engineer May 2026 – Present

Data Analyst, Supply Chain Sep 2025 – May 2026

- **Fully recovered $768.8K in IEEPA duty refunds** by independently leading TCI's end-to-end Phase I recovery initiative: extracted and reconciled ACE data, screened 98 entries across two importer entities, and validated duty-line classifications.
- Promoted on the strength of that recovery to lead TCI's AI automation initiative (AI TOP), partnering with every operation department to map operational pain points and automate their workflows. Built a centralized dashboard integrating shipment-tracking APIs with procurement and logistics systems for freight-cost, delivery-time, and exception monitoring.
- Built and owned production ETL pipelines from SAP ERP into Databricks, modeling shipment, inventory, warehouse-order, and multi-tier vendor data into auto-refreshed dashboards. Improved reporting efficiency 40%.
- Manage end-to-end inbound and outbound international trade logistics while supporting company-wide procurement, applying data and automation to purchasing and logistics decisions.

First Solar *Perrysburg, OH*

Data Scientist, R&D Data Quality Sep 2024 – Jan 2025

- Led validation of data quality across metrology tools in solar module production, using statistical process control and root cause analysis to identify and resolve anomalies.
- Built automated data pipelines using Python, SQL, and scheduled batch scripts running every 20 minutes to validate metrology datasets in near real-time, reducing manual review time and increasing the reliability of quality metrics. Developed custom JMP scripts to extract and clean data, effectively filtering out production noise and decreasing false alarm rates by 80%.
- Created real-time KPI dashboards in JMP Live to monitor performance across 30+ manufacturing tools, accelerating issue response and reducing troubleshooting time by 50% while supporting continuous-improvement and Lean Six Sigma initiatives.

Sanofi (China) Investment Co., Ltd. *Shanghai, China*

Data Analyst Jun 2022 – Sep 2022

- Designed and validated SQL/MongoDB ETL pipelines standardizing and loading **10M+ rows** of patient-demand data for consistent enterprise reporting.
- Ran data-integrity checks and validated business logic for drug-access and demand-forecasting programs alongside supply-chain stakeholders.

PROJECT

PO & Logistics Tracker | *Internal self-service web app (SQL, Python, Databricks)*

- Developed a logistics and warehouse operations system linking purchase orders with domestic, import, and export shipments, providing real-time visibility into shipment status, warehouse receiving, and required actions while reducing manual freight-rate comparison time by **70%.**

RESEARCH

- Modeled global trade and R&D-investment dynamics with econometric methods (VAR and two-way fixed-effects gravity models) across 31 provinces, 2001 to 2021, as a graduate research assistant. Tools: Stata, Python.

EDUCATION

University of Wisconsin–Madison M.S. in Information, GPA 3.93 Madison, WI

Xi'an Jiaotong-Liverpool University B.S. in Economics & Finance, GPA 3.8 Suzhou, China